

The first term is manifestly covariant, and the second term, though not covariant, is local, so it can be cancelled by adding a local non-covariant term to the Hamiltonian density. Specifically, if $V_\mu(x)$ interacts with other fields through a term $V_\mu(x)J^\mu(x)$ in $\mathcal{H}(x)$, then the effect of the second term in Eq. (6.2.21) is to produce an effective interaction

$$-i\mathcal{H}_{eff}(x) = \frac{i}{2} \left[-iJ^\mu(x) \right] \left[-iJ^\nu(x) \right] \left[-im^{-2}\delta_\mu^0\delta_\nu^0 \right].$$

(The factors $-i$ are the usual ones which always accompany vertices and propagators. The factor $\frac{i}{2}$ is needed because there are two ways to pair other fields with $\mathcal{H}_{eff}(x)$, differing in the interchange of J^μ and J^ν .) Thus the effect of the non-covariant second term in Eq. (6.2.21) can be cancelled by adding to $\mathcal{H}(x)$ the non-covariant term

$$\mathcal{H}_{NC}(x) = -\mathcal{H}_{eff}(x) = \frac{1}{2m^2} \left[J^0(x) \right]^2. \quad (6.2.22)$$

It is the singularity of the equal-time commutators of vector fields at zero separation that requires us to employ a wider class of interactions than those with a scalar density. A detailed non-perturbative proof of the Lorentz invariance of the S -matrix in this theory will be given in the next chapter.

It should not be thought that this is solely a phenomenon associated with spins $j \geq 1$. For instance, consider the vector field associated with a particle of spin $j = 0$, equal (as discussed in Chapter 5) to the derivative $\partial_\lambda\phi(x)$ of a scalar field. For the pairing of this field with a scalar $\phi^\dagger(y)$, the polynomial $P(p)$ on the mass shell is

$$P_\lambda(p) = ip_\lambda, \quad (6.2.23)$$

while the pairing of $\partial_\lambda\phi(x)$ with $\partial_\eta\phi^\dagger(y)$ yields a polynomial

$$P_{\lambda,\eta}(p) = p_\lambda p_\eta. \quad (6.2.24)$$

The covariant polynomials for general off-shell four-momenta q^μ are again obtained by just substituting q^μ for p^μ in Eqs. (6.2.23) and (6.2.24). Eq. (6.2.23) shows that $P_\lambda(q)$ is already linear in q_0 , so here there is no difference between $P_\lambda(q)$ and $P_\lambda^{(L)}(q)$. However, for Eq. (6.2.24) there is a difference:

$$\begin{aligned} P_{\lambda,\eta}^{(L)}(q) &= q_\lambda q_\eta - (q_0^2 - \mathbf{q}^2 - m^2)\delta_\lambda^0\delta_\eta^0 \\ &= P_{\lambda,\eta}(q) + (q^2 + m^2)\delta_\lambda^0\delta_\eta^0, \end{aligned} \quad (6.2.25)$$

so here the propagator is

$$\Delta_{\lambda,\eta}(x, y) = (2\pi)^{-4} \int d^4q \frac{q_\lambda q_\eta e^{iq \cdot x}}{q^2 + m^2 - i\epsilon} + \delta_\lambda^0\delta_\eta^0\delta^4(x - y). \quad (6.2.26)$$